SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	**T.Rajashree**	**Reg. No.:**	**192472393**
Section / Batch:	**CSE-AI**	**Date:**	**10-10-26**

Code of Conduct

I T.Rajashree(192472393)_____ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

7. Construct a concept map that evaluates the operational efficiency of a Public Land Mobile Network (PLMN).
   Relate operators, subscribers, switching centers, and radio access networks.
   Assess how these components collectively provide seamless connectivity.
   Justify the relationships included in the concept map.

Justify your recommendation through well-defined conceptual relationships.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem Solving Approach	20	Logical, well structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



- **Operator and Subscriber Coordination**
  - Operator Governance
    - Owns spectrum, sites, core network equipment, service platforms, and operating policies
    - Monitors alarms, key performance indicators, traffic trends, and faults
    - Balances coverage, capacity, quality, resilience, and operating cost
  - Subscriber Experience
    - Generates voice, messaging, data, mobility, and emergency-service demand
    - Expects availability, throughput, low latency, call quality, and successful handovers
    - Moves between cells and service areas while expecting session continuity
  - Demand and Resource Alignment
    - Operator investment follows subscriber traffic and mobility patterns
    - Network configuration adapts to congestion, coverage gaps, faults, and changing demand
    - Efficient coordination prevents underused capacity and overloaded network elements
- **Radio Access and Mobility**
  - Wireless Access
    - Subscriber device requests registration or service through the nearest suitable cell
    - Cells, base stations, antennas, and radio controllers provide the air interface
    - Radio measurements identify signal conditions and suitable serving cells
  - Radio Resource Management
    - Allocates radio resources to active subscribers
    - Applies scheduling and spectrum reuse to serve more demand
    - Controls coverage, interference, prioritization, and radio-link quality
  - Mobility Management
    - Cell selection and reselection maintain an appropriate serving cell
    - Handover procedures transfer an active connection as radio conditions or location change
    - Switching centers preserve subscriber identity and session control across changing cells
  - **Seamless Radio Continuity**
    - Timely measurements and handover coordination reduce dropped sessions
    - The radio access network maintains the wireless link while the core maintains service context
    - Inter-system mobility supports roaming between different network technologies
- **Switching Centers and Service Routing**
  - Subscriber and Session Control
    - Authenticates subscribers and establishes mobility and session context
    - Coordinates location updates, paging, handovers, interconnection, and charging
    - Scales signaling, authentication, routing, and session-handling capacity
  - Traffic Routing
    - Receives traffic from the radio access network through transport facilities
    - Routes calls, messages, and packet-data sessions to the correct destination
    - Connects subscribers to other subscribers, external networks, or service platforms
  - Service Authorization
    - Applies operator policies for service authorization and resource priorities
    - Maintains subscriber identity and session state beyond the serving cell
    - Supports policy enforcement, resilience, and service-quality management
  - Core-Radio Interdependence
    - Radio access networks need switching centers for authentication, mobility control, and routing
    - Switching centers need radio access networks to provide the physical wireless path to subscribers
    - Together they transform local radio access into end-to-end mobile communication

Operational Efficiency and Seamless Connectivity	
Coverage and Availability	Radio access coverage supports service throughout the operating area
	Distributed cells improve access and contain localized failures
	Switching-center redundancy limits the impact of core-network failures
	Operator maintenance and fault response restore service quickly
Capacity and Resource Use	Radio scheduling and spectrum reuse increase use of available radio resources
	Switching capacity supports signaling, authentication, routing, and session handling
	Operators match capacity and configuration to subscriber traffic demand
Quality and Responsiveness	Subscribers receive stable calls, adequate throughput, and predictable latency
	Radio measurements enable timely handovers and reduce radio-link failures
	Efficient signaling and routing reduce service delay
Resilience and Scalability	Redundant core functions and distributed radio sites improve fault tolerance
	Operators add capacity, sites, spectrum, or core resources as demand grows
	Interworking and roaming extend service continuity beyond one operator domain
Collective Connectivity Result	Operators provide governance, infrastructure, resources, and monitoring
	Subscribers create demand and mobility events
	Radio access networks provide local wireless connectivity
	Switching centers provide identity, mobility, routing, and session continuity
	A failure or bottleneck at any interface degrades the end-to-end subscriber experience

Presented with xmind